

Math 344, F18
Quiz 2

Name: Answer

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (6 points) Given two matrices $A = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & -4 & 0 \\ 1 & 3 & 1 \end{bmatrix}$.

Compute AB and BA .

$$\begin{aligned} AB &= \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & -4 & 0 \\ 1 & 3 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 2 & 3 \\ 0 & -8 & -6 \\ 0 & 1 & -2 \end{bmatrix} \\ BA &= \begin{bmatrix} 1 & 2 & 3 \\ 2 & -4 & 0 \\ 1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -6 & 2 & 3 \\ 10 & -4 & 0 \\ -6 & 3 & 1 \end{bmatrix} \end{aligned}$$

Problem 2. (4 points) Write down the 3×3 matrices that produce the following operations:

(a) E_{21} subtracts $1/2$ times row 1 from row 2.

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(b) E_{32} adds 7 times row 2 to row 3.

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 7 & 1 \end{bmatrix}$$